



NYU

Blast Tube Model Final Presentation

17 April 2025

PRESENTED BY Jing Yu Carolina Cen Feng, Saba
Khan

Project Final Submission

Design Criteria

1. Area of the device is less than or equal 20% of the blast tube cross sectional area.
2. Quick insertion of the PDMS substrate inside the apparatus (eliminate bolts).
3. Elimination of all air bubbles.
4. How the apparatus will be mounted inside the blast tube.
5. Fitting the PDMS substrate inside a commercially available petri dish.
6. Select Material for all you design components that are biocompatible and autoclavable.
7. Pressure sensor integration.

CBA Design and Assembly

01

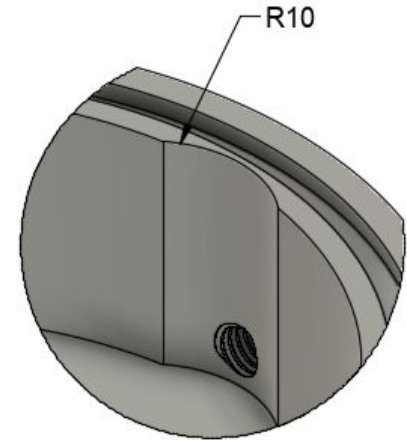
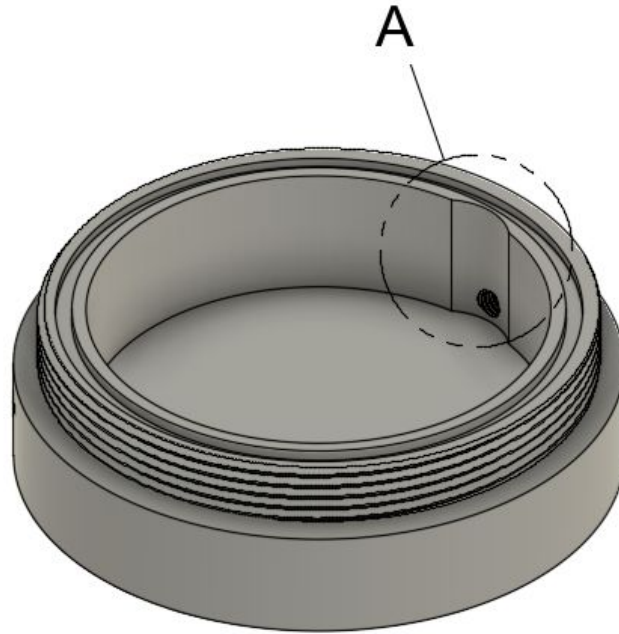
CBA Bottom



CBA Bottom

Indentation
made at outlet
hole to allow for
easy release of air
bubbles.

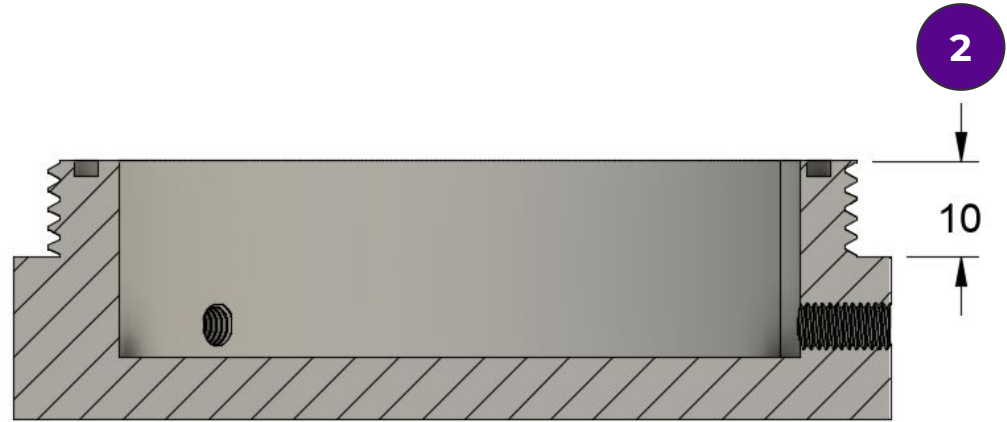
✓ Elimination
of Air Bubbles



DETAIL A
SCALE 2:1

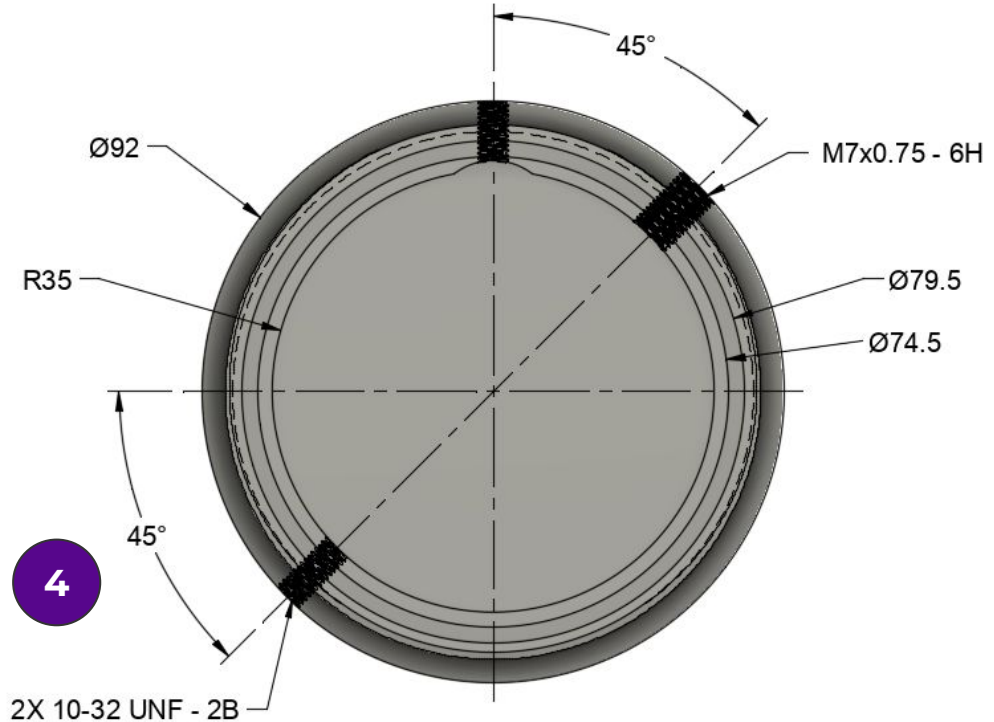
CBA Bottom - Revisions

- 1. Material Change:**
HDPE →
Polyphenylsulfone (PPSU)
- 2. Increased threaded
portion height:**
8mm → 10mm
- 3. Scaled Overall Size
Down:**
1:1 → 1:0.9 → 1:0.75



CBA Bottom - Revisions

4. Inlet hole positioned moved:
45 degrees CCW



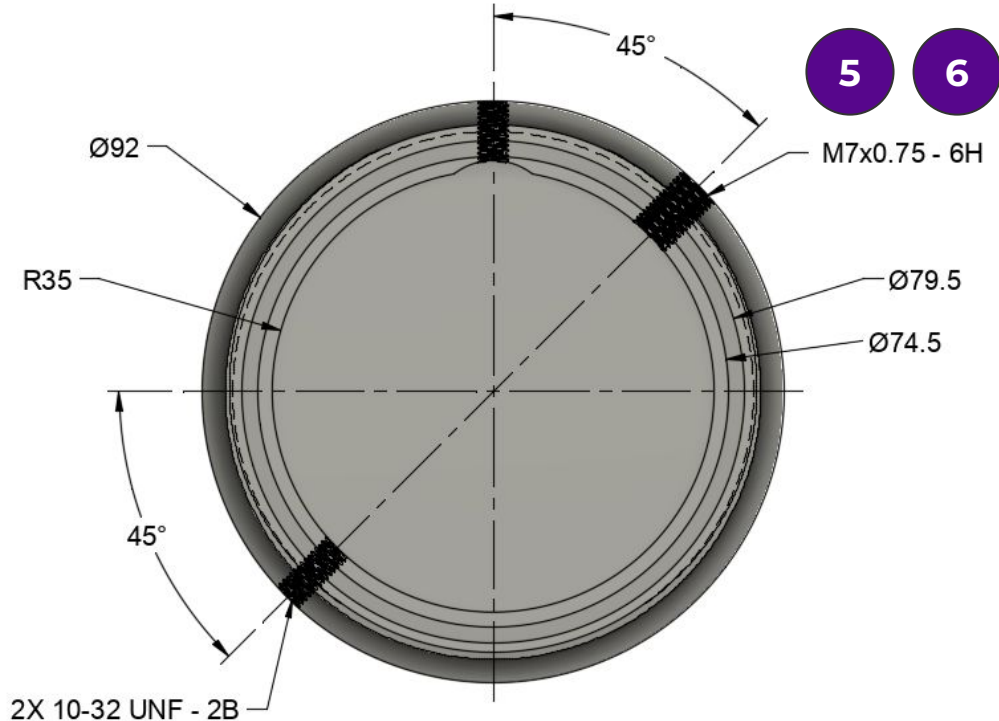
CBA Bottom - Revisions

5. Pressure sensor hole position moved:

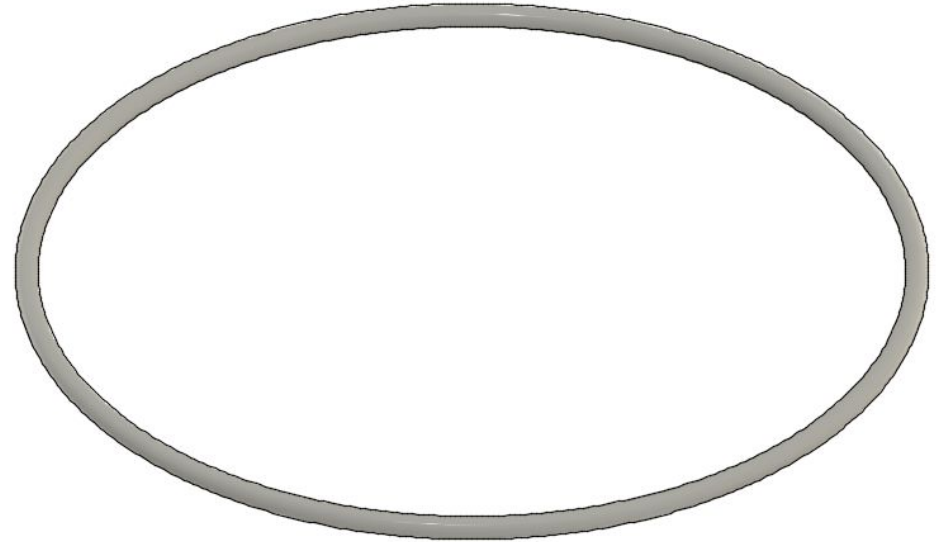
45 degrees CW

6. Threads added:
M7x0.75 - 6H

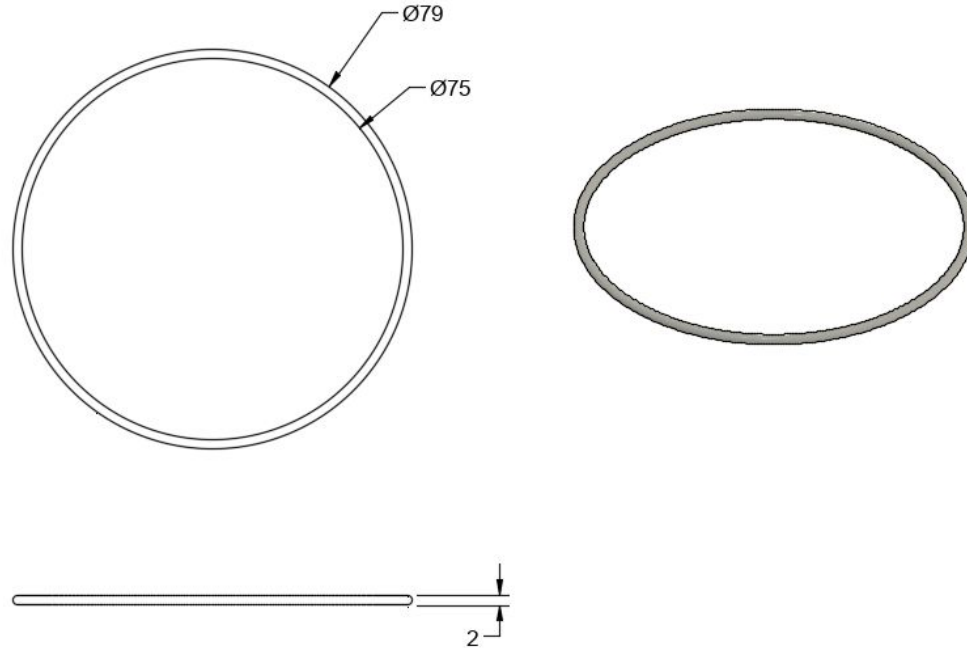
✓ Pressure sensor integration



O-Ring



Selected O-Ring



Design purpose: O-Ring
Insertion to ensure
Watertight seal

Material: Buna-N, oil
resistant

Revisions Made:

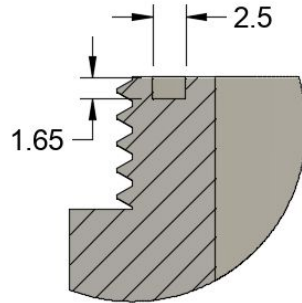
- ID: 111mm $\rightarrow$ 100mm $\rightarrow$ 75 mm
- OD: 117mm $\rightarrow$ 105mm $\rightarrow$ 79 mm

O-Ring Insertion Dimensions

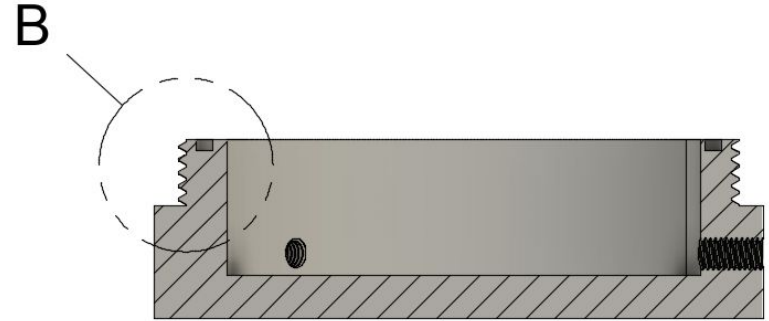
O-Ring Width:
2 mm

Groove Depth:
1.65 mm

Groove Width:
2.5 mm

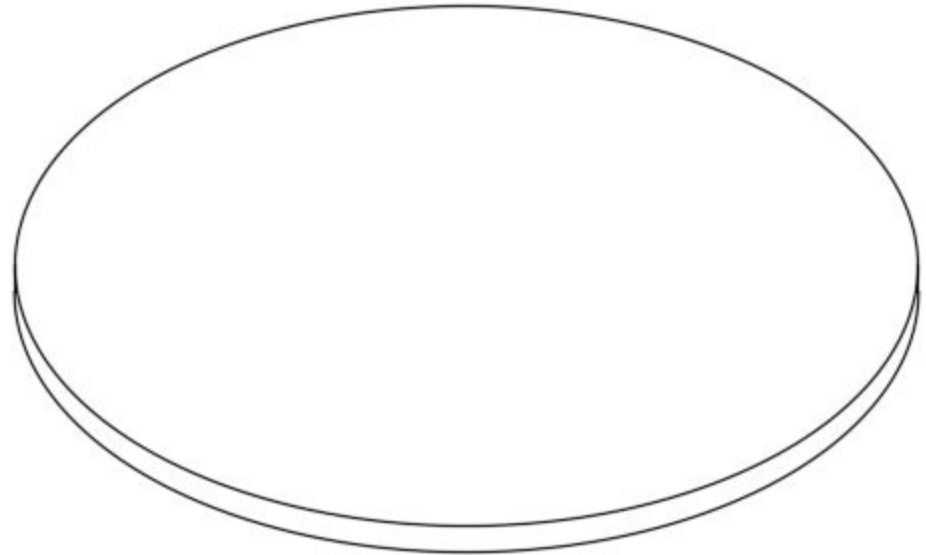


DETAIL B
SCALE 2:1



SECTION A-A
SCALE 1:1

Acrylic Top Cover



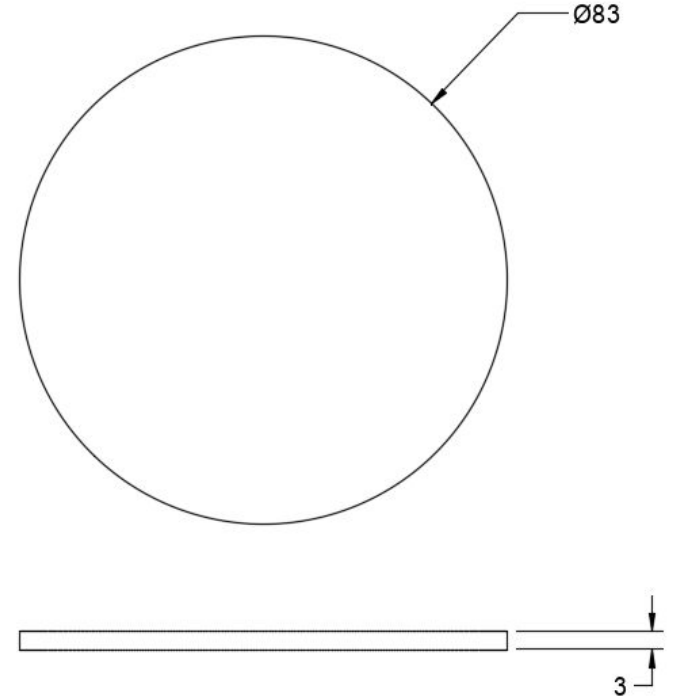
Clear Acrylic Disk

Design and purpose: act as a lid covering the PDMS substrate and the lid apparatus

Material: Clear Polycarbonate

Revisions Made:

- Diameter: 124.5 mm → 112.05 mm → 83 mm
- Thickness: 3.00 mm → Unchanged → Varies from 3mm to 8mm



CBA Rim



CBA Rim - Revisions

1. Material Change:

HDPE →

Polyphenylsulfone (PPSU)

2. Increased threaded portion height:

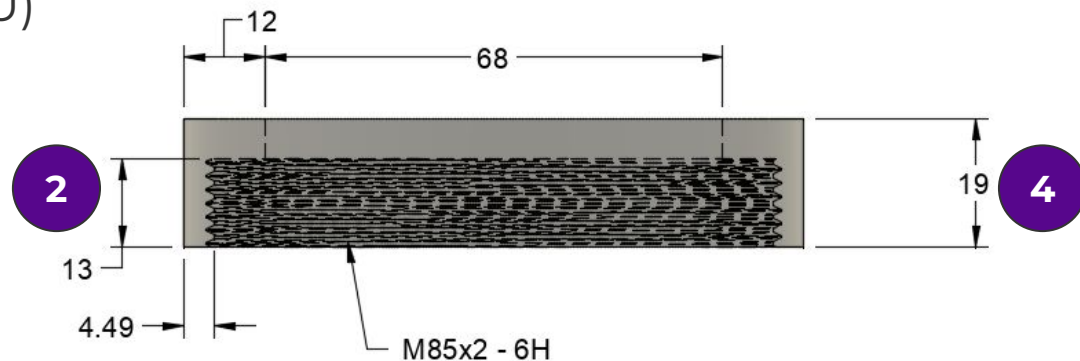
11mm → 13mm

3. Scaled Overall Size Down:

1:1 → 1:0.9 → 1:0.75

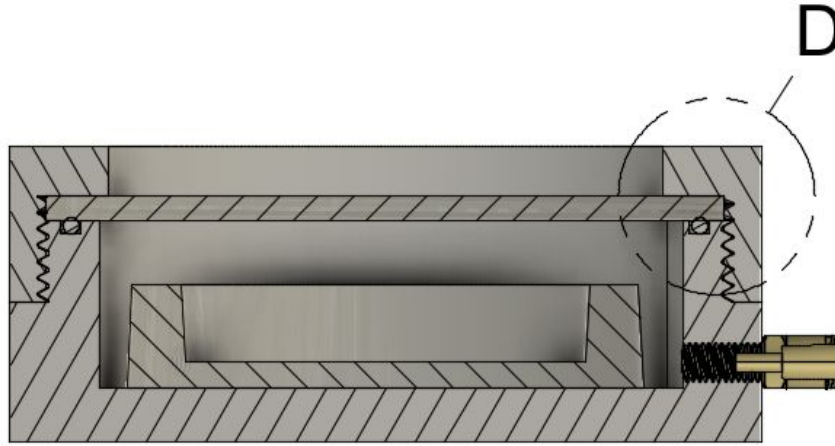
4. Increased thickness:

17mm → 19mm

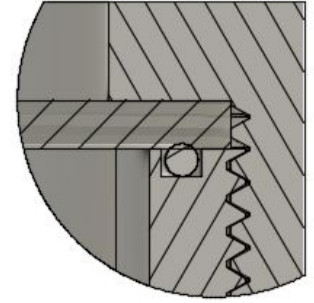


Cross Sectional View

3mm
Thick
Acrylic



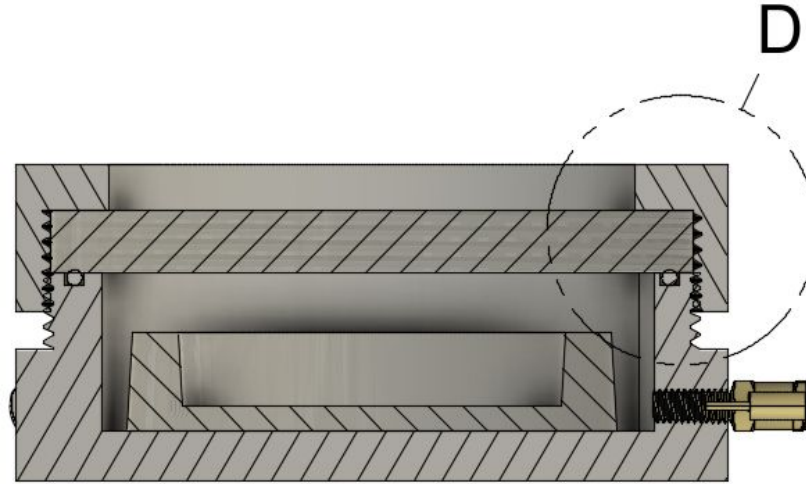
SECTION C-C
SCALE 1:1



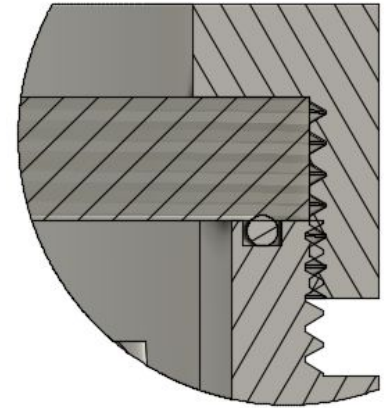
DETAIL D
SCALE 2:1

Cross Sectional View

8mm
Thick
Acrylic



SECTION C-C
SCALE 1:1



DETAIL D
SCALE 2:1

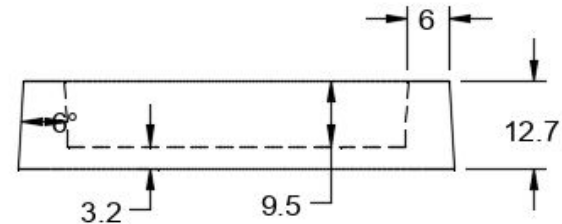
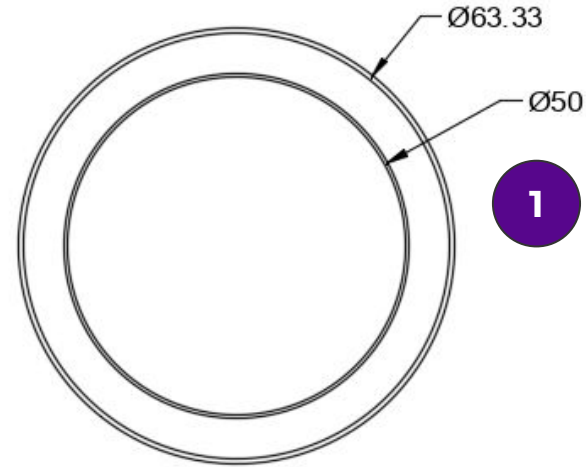
PDMS Substrate

1. Decreased Overall Size:

Example:

OD: 75mm $\rightarrow$ 63.33mm

ID: 60.42mm $\rightarrow$ 50 mm

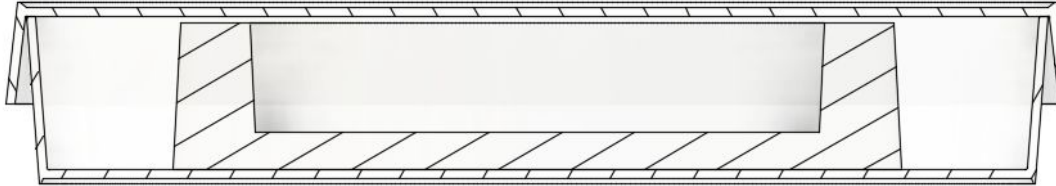


PDMS Inside Petri Dish

*100 mm
Diameter
Petri Dish*



*McMaster
Carr*



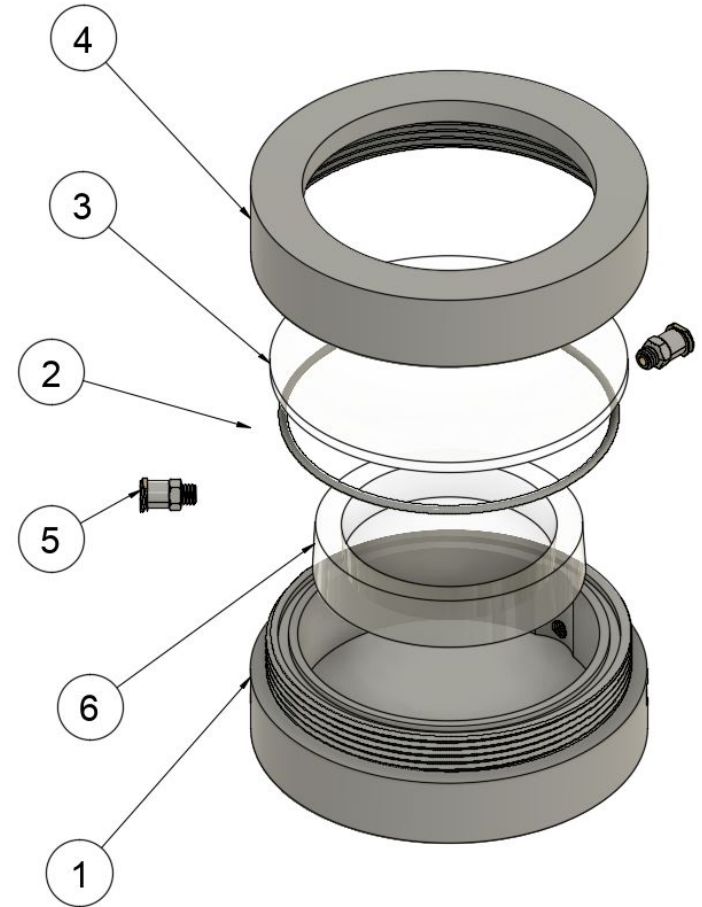
PDMS Fits Inside Commercially Available Petri Dish

CBA Assembly

Parts List

ITEM	QTY	NAME	MATERIAL
1	1	CBA BOTTOM	PPSU (POLYPHENYLSULFONE)
2	1	O-RING	BUTYL RUBBER
3	1	ACRYLIC TOP COVER	CLEAR POLYCARBONATE
4	1	CBA RIM	PPSU (POLYPHENYLSULFONE)
5	2	LUER INSERT	POLYCARBONATE PLASTIC
6	1	PDMS	PDMS

✓ All material is autoclavable and biocompatible.



Most Critical Tolerances

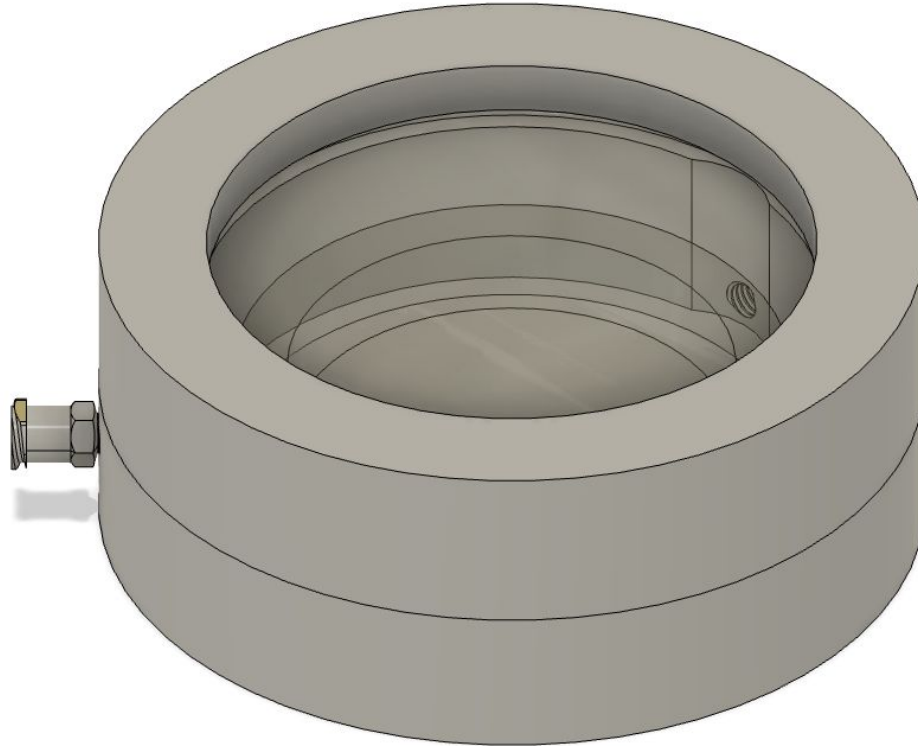
1. O-Ring Groove on CBA Bottom

- a. **Features:** ID, Width, Depth, Surface Finish
- b. **Reason:** Ensure watertight seal and avoid leakage

2. CBA Bottom and Rim Threaded Interface

- a. **Features:** Thread Depth, Concentricity
- b. **Reason:** Misalignment would cause leaks or assembly failure

Cell Blast Apparatus



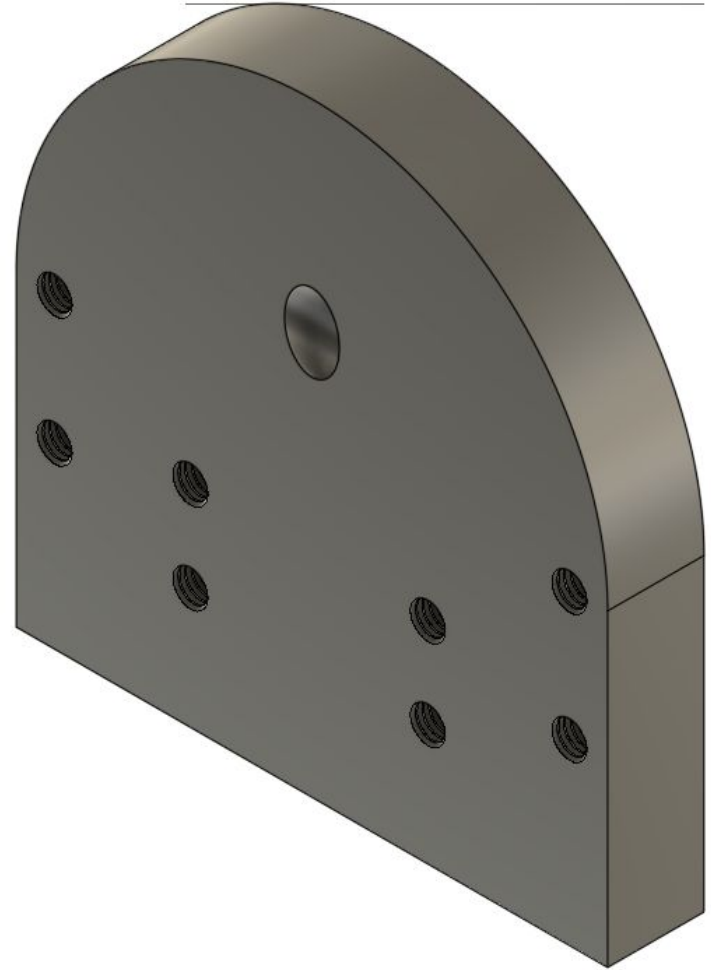
Design Criteria

1. Area of the device is less than or equal 20% of the blast tube cross sectional area.
2. Quick insertion of the PDMS substrate inside the apparatus (eliminate bolts). ✓
3. Elimination of all air bubbles. ✓
4. How the apparatus will be mounted inside the blast tube.
5. Fitting the PDMS substrate inside a commercially available petri dish. ✓
6. Select Material for all you design components that are biocompatible and autoclavable. ✓
7. Pressure sensor integration. ✓

Mounting Apparatus (MA) Design and Assembly

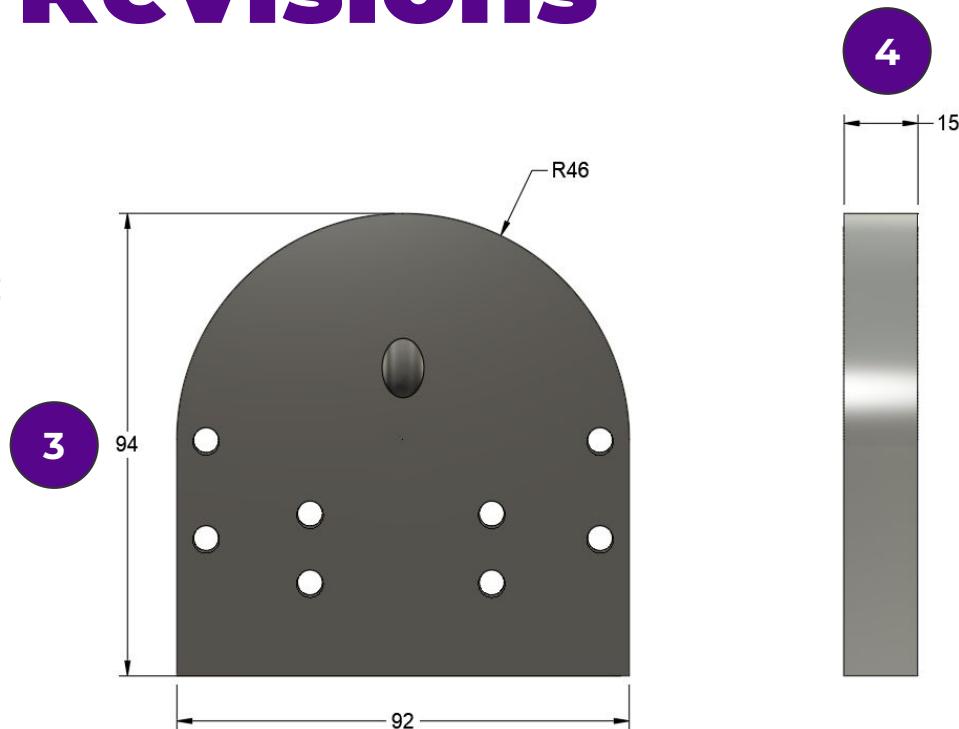
02

Headboard



Headboard - Revisions

- 1. Material Specified:**
Steel AISI 1045 450 QT
- 2. Scaled Overall Size Down:**
1:1 $\rightarrow$ 1:0.75
- 3. Decreased height:**
119.169mm $\rightarrow$ 94mm
- 4. Increased thickness:**
9mm $\rightarrow$ 15mm



Headboard - Revisions

5. Added holes and threads:

M6 x 1mm (x8) -

Brackets

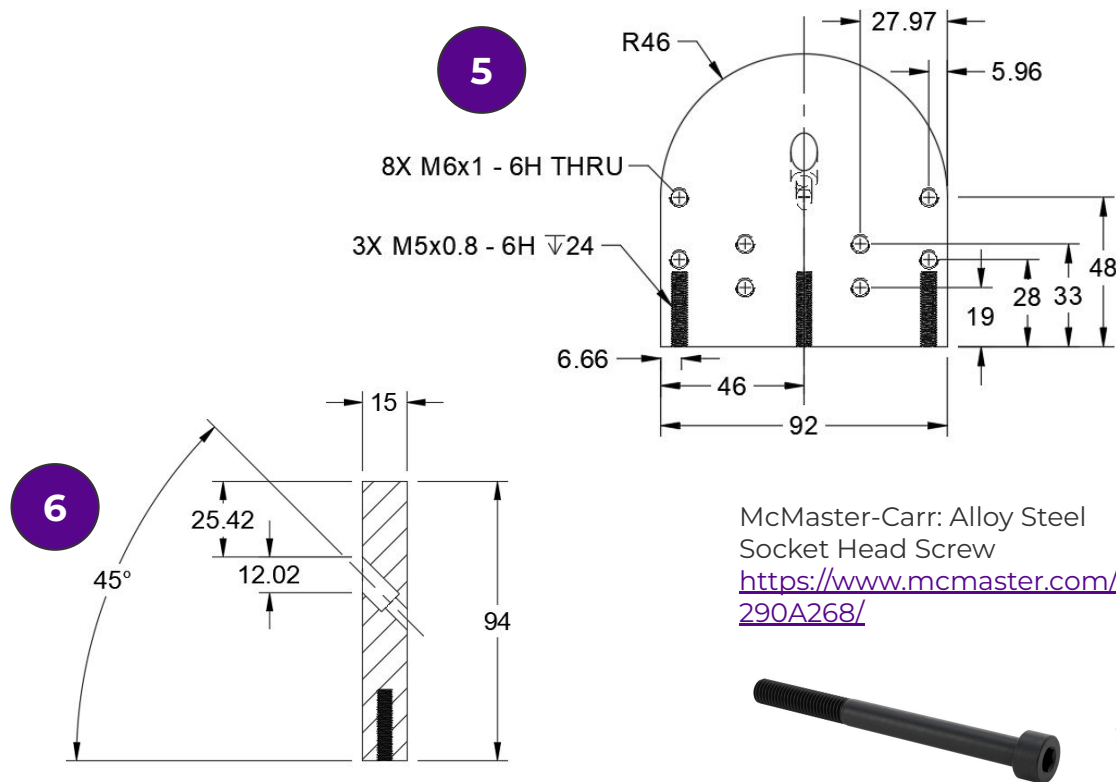
M5 x 0.8mm (x3) -

Baseboard

Attachment

6. Added 45° Counterbore hole

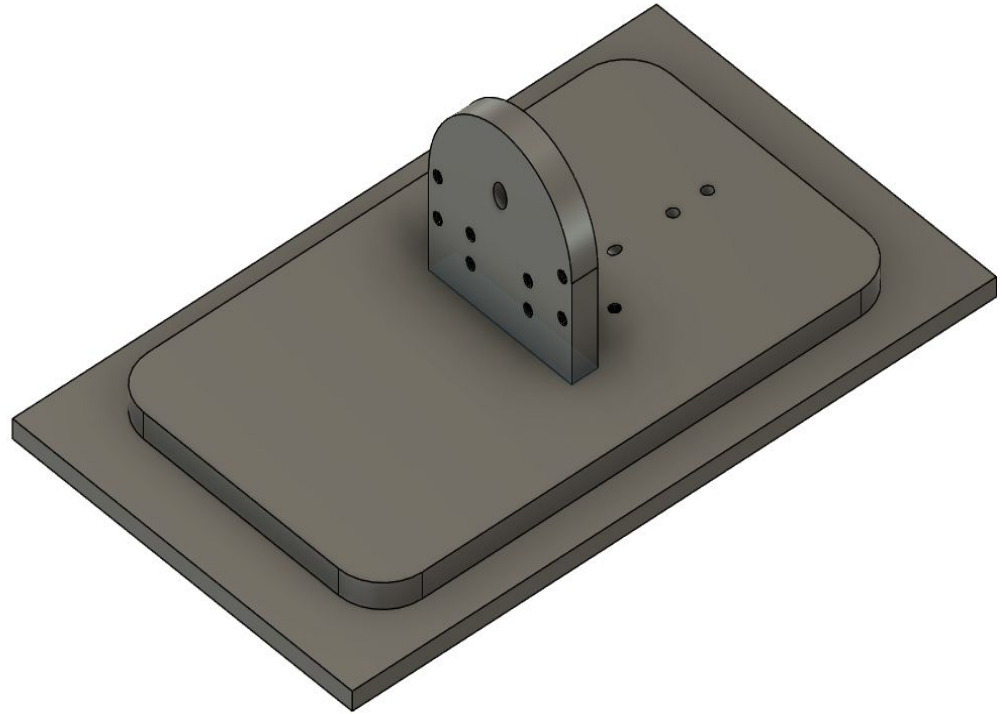
Clearance hole -
T-Slot



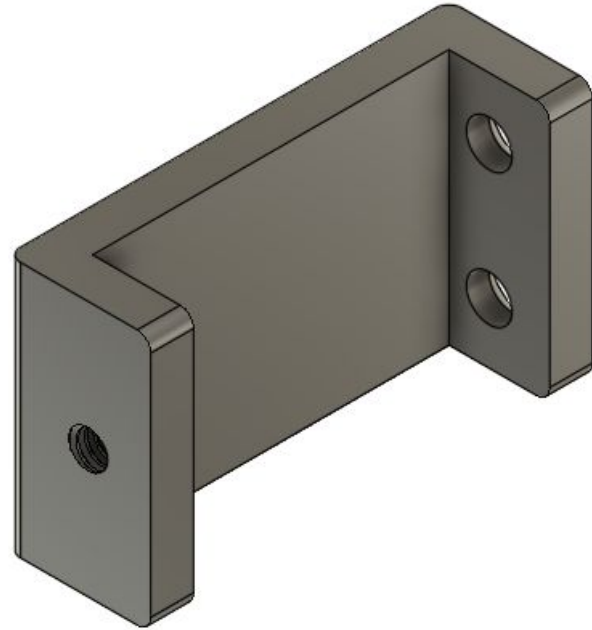
McMaster-Carr: Alloy Steel
Socket Head Screw
<https://www.mcmaster.com/91290A268/>



Headboard on the Bottom Plate

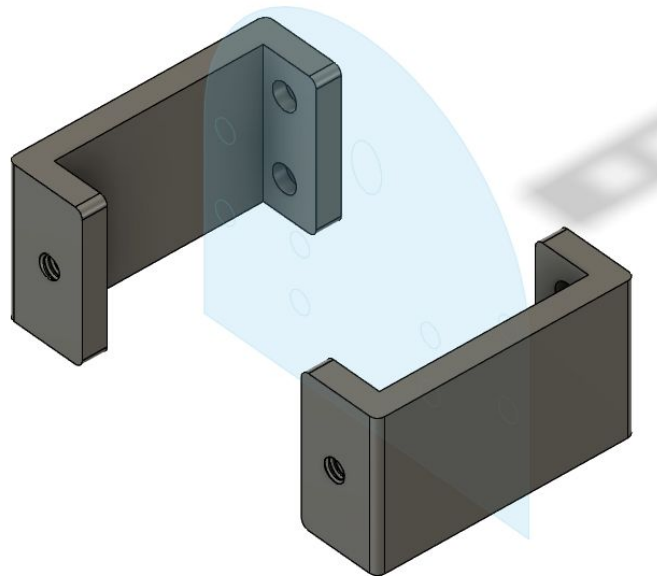


C-Brackets (x2)



C-Brackets - Revisions

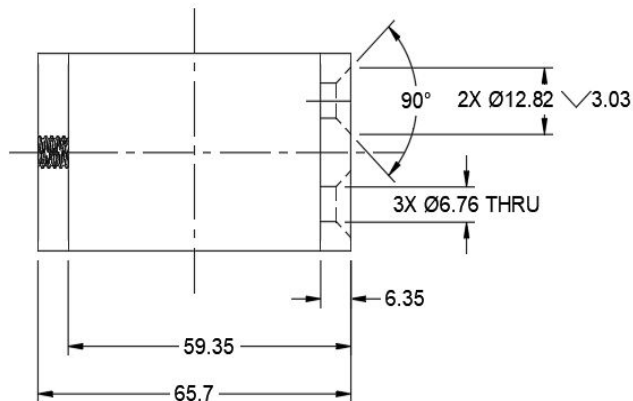
- 1. Material Specified:**
Steel AISI 1045 450 QT
- 2. Scaled Overall Size Down:**
1:1 $\rightarrow$ 1:0.75
- 3. Gap elongated/wider:**
50.80mm $\rightarrow$ 59.35mm



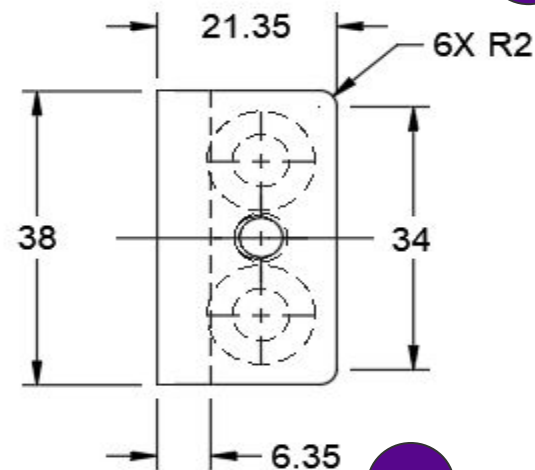
C-Brackets - Revisions

5

- 4. Increased thickness:
5.08mm → 6.35mm
- 5. Sharp edges removed:
2mm radius fillet
- 6. 2X Clearance + 90° countersink holes
- 7. 1X threaded hole



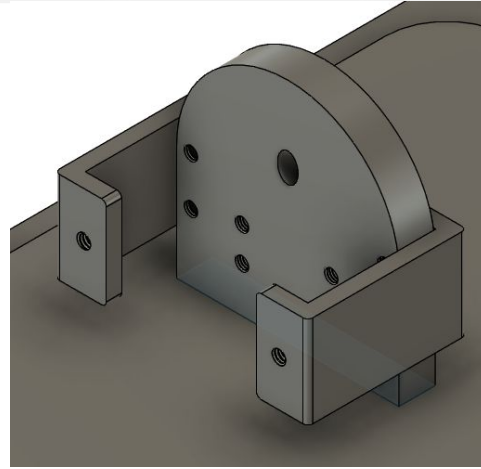
6



4

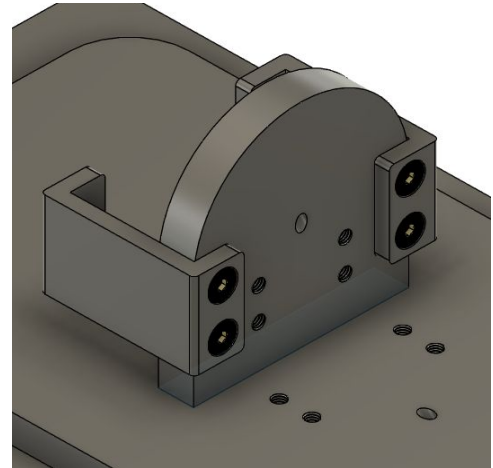
C-brackets added to the headboard

McMaster-Carr: Black-Oxide Alloy
Steel Hex Drive Flat Head Screw
<https://www.mcmaster.com/91294A240/>

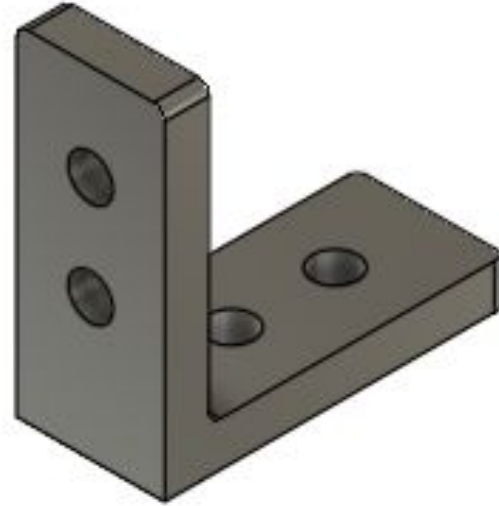


*Auxiliary
Front View*

*Auxiliary
Back View*

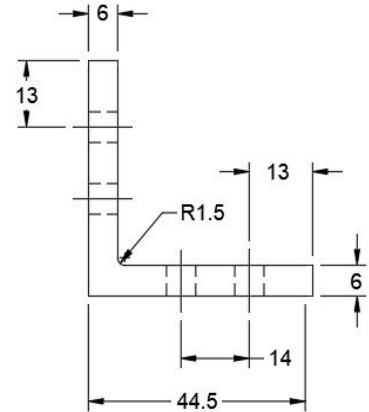
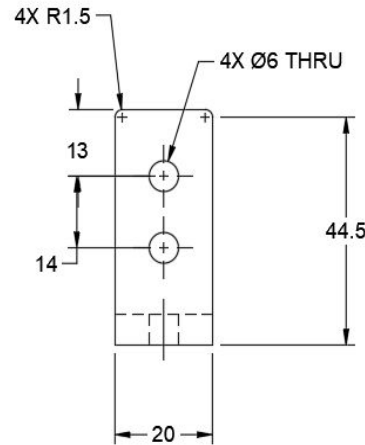


90° Bracket (x2)



90° Bracket - Addition

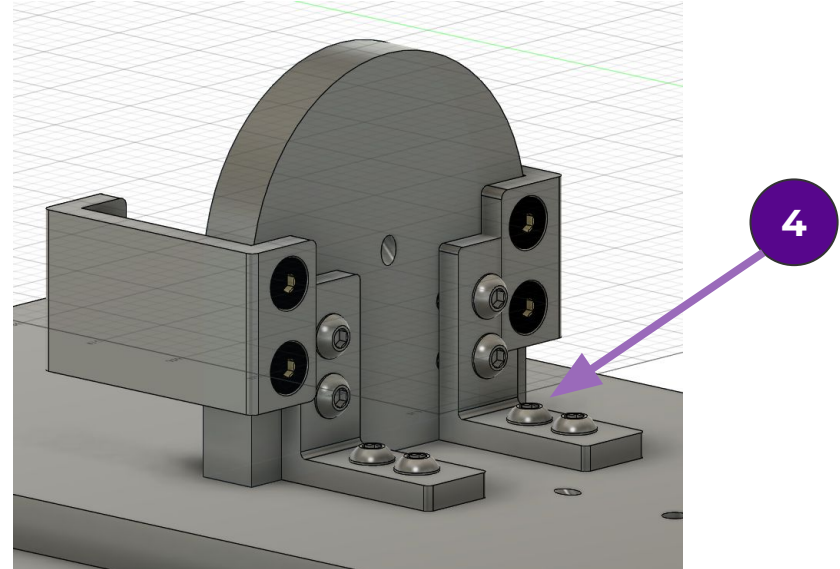
- 1. Material Specified:**
Steel AISI 1045 450 QT
- 2. Purpose:**
Provide additional support for the headboard
- 3. Clearance hole used:**
4x 6mm through holes for each bracket



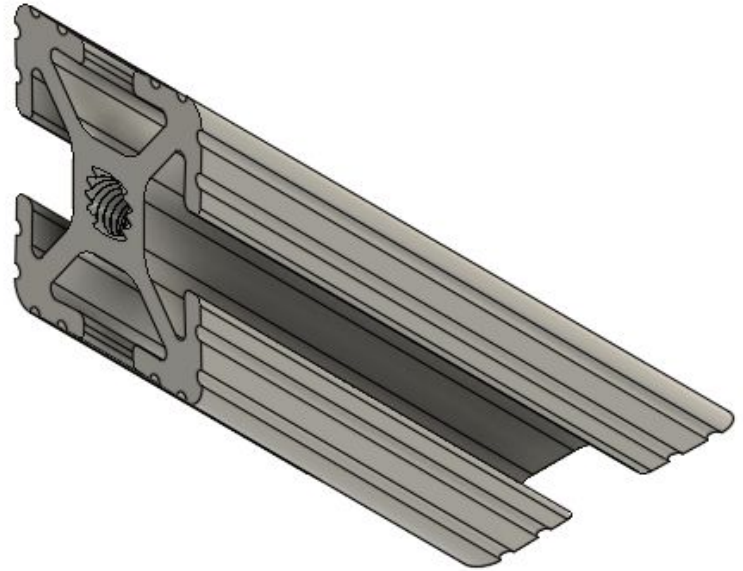
90° Bracket - Addition

- 4. Screws added:**
Passivated 18-8 Stainless Steel, M6 x 1 mm Thread, 14mm Long

From McMaster-Carr: Button Head Hex Drive Screw
<https://www.mcmaster.com/92095A227/>

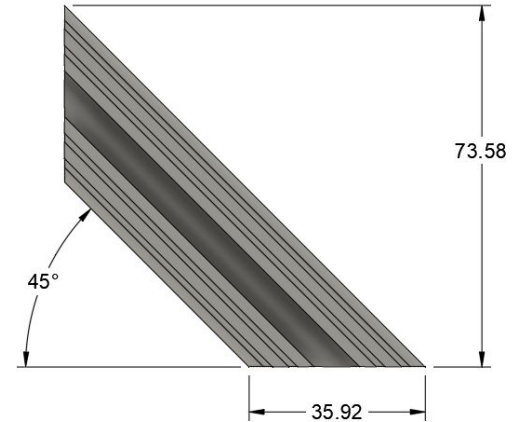
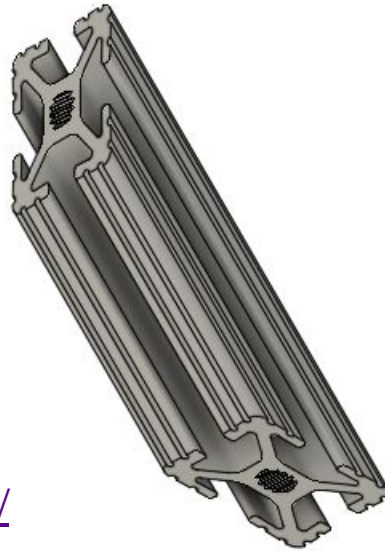


45° T-Slot



45° T-Slot - Revisions

1. **Shortened length:**
102.79mm → 73.58mm
2. **Material:**
6560 Aluminium

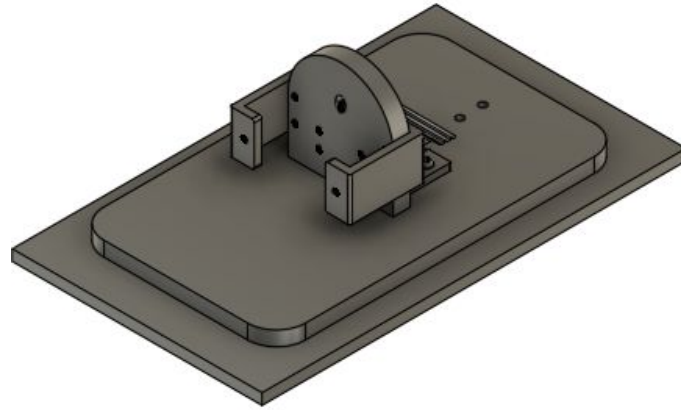


From McMaster-Carr: T-Slotted
Framing Rail
<https://www.mcmaster.com/4601N24/>

McMaster-Carr: Alloy Steel
Socket Head Screw
<https://www.mcmaster.com/91290A268/>

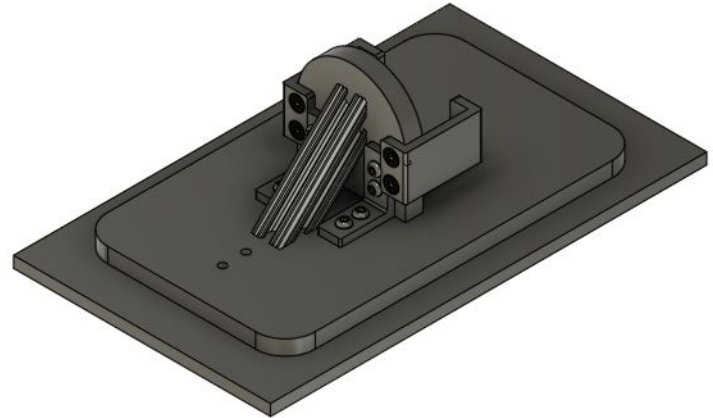


Integration of the 45° T-Slot

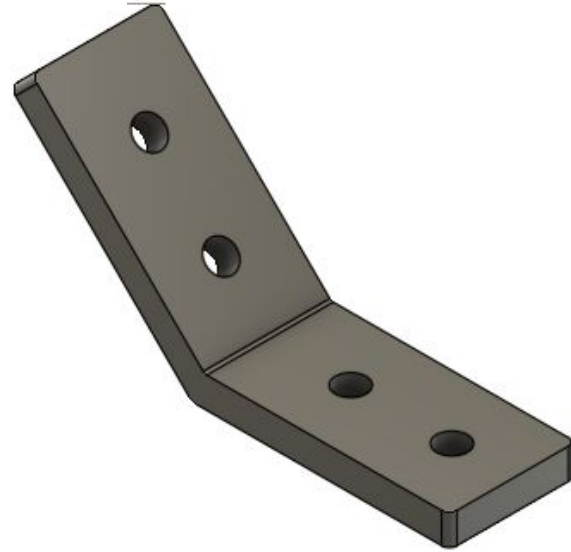


*Auxiliary
Front View*

*Auxiliary
Back View*

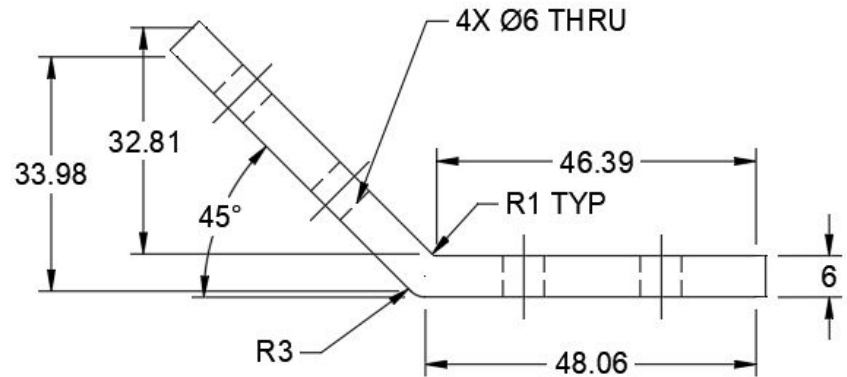


45° Bracket



45° Bracket - Addition

- 1. Material Specified:**
Steel AISI 1045 450 QT
- 2. Purpose:**
Provide additional support for the 45° T-Slot
- 3. Clearance hole used:**
4x 6mm through holes



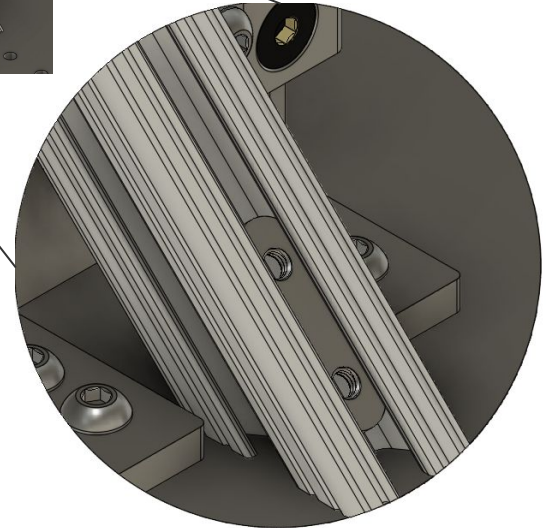
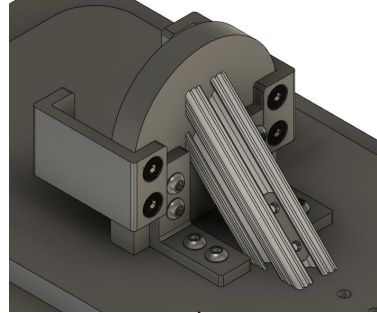
45° Bracket - Addition

- 4. T-slot Framing added:**
End-Feed Double Nut and
M5 Flanged Button Head,
20mm Rail

From McMaster-Carr:

<https://www.mcmaster.com/5537T46>

4/



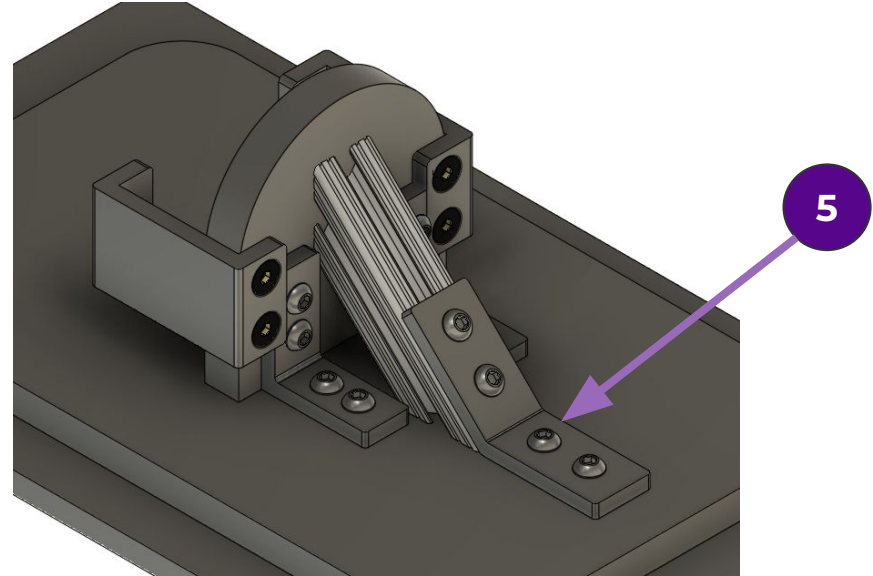
45° Bracket - Addition

5. Screws added:

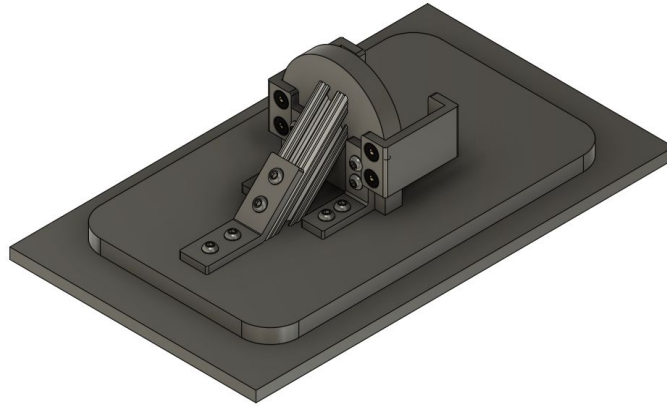
Passivated 18-8 Stainless Steel, M6 x 1 mm Thread, 14mm Long

From McMaster-Carr:

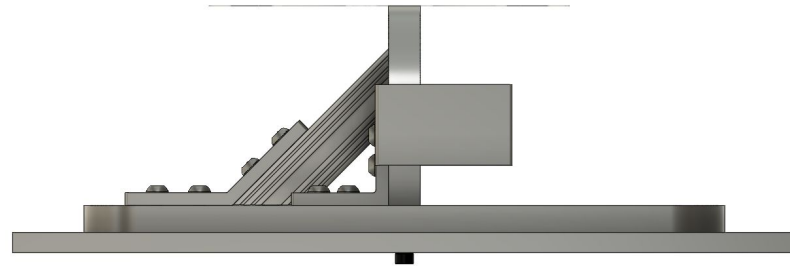
<https://www.mcmaster.com/92095A227/>



*Placing
the
45° bracket
to
45° T-Slot*



*Auxiliary
Back View*



Side View

Soft-tip Thumb Screw



Soft-tip Thumb Screw

1. Material:

Tip: Nylon plastic

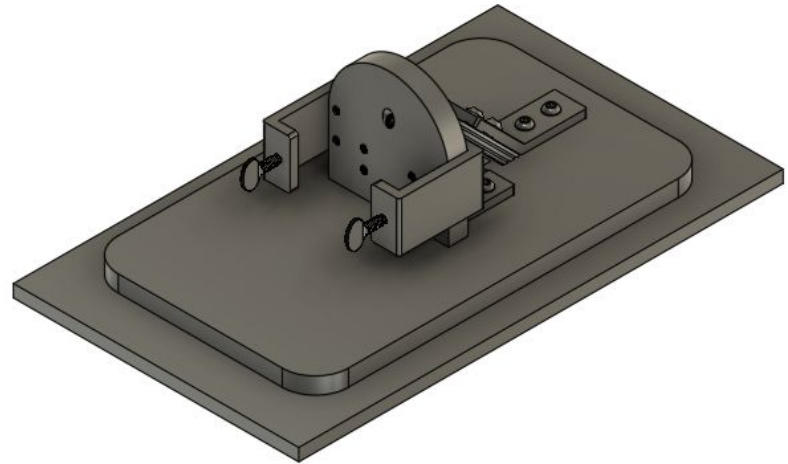
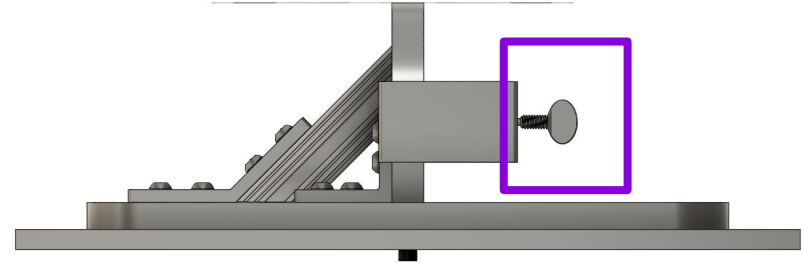
Screw body: Zinc-Plated Steel

2. Purpose:

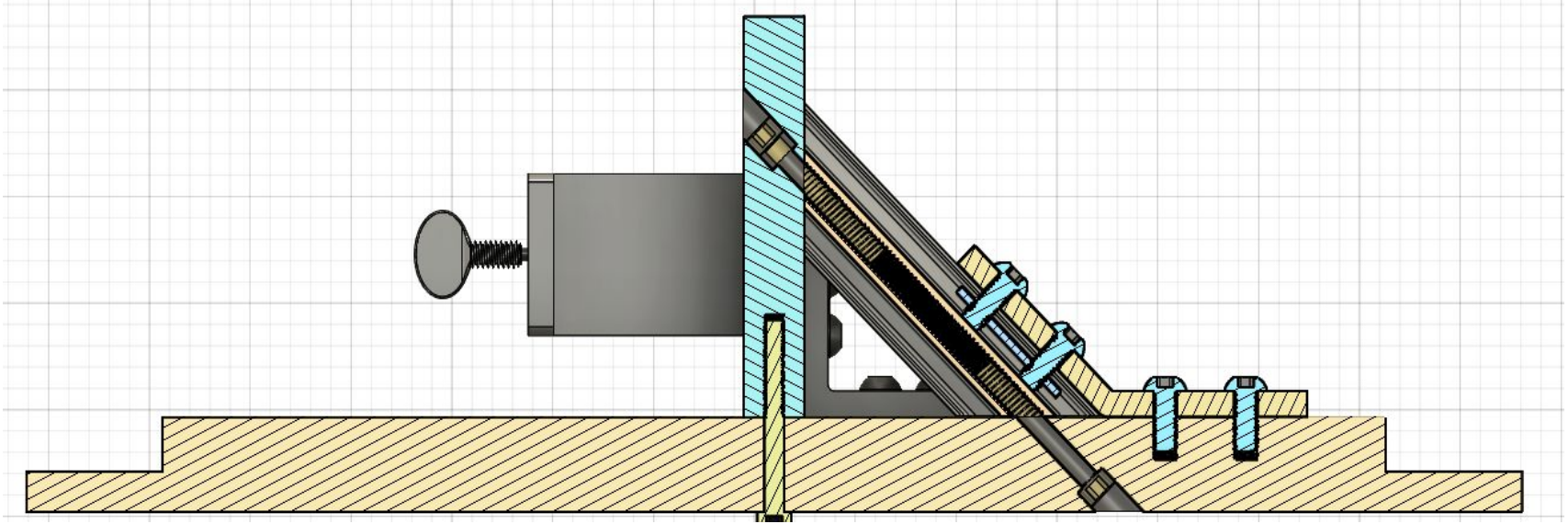
Secure the CBA onto the headboard

From McMaster-Carr:

<https://www.mcmaster.com/90772A537/>

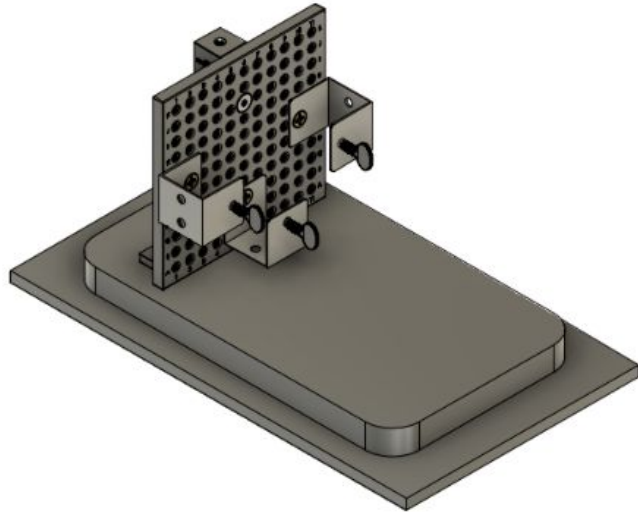


Cross-Section Views

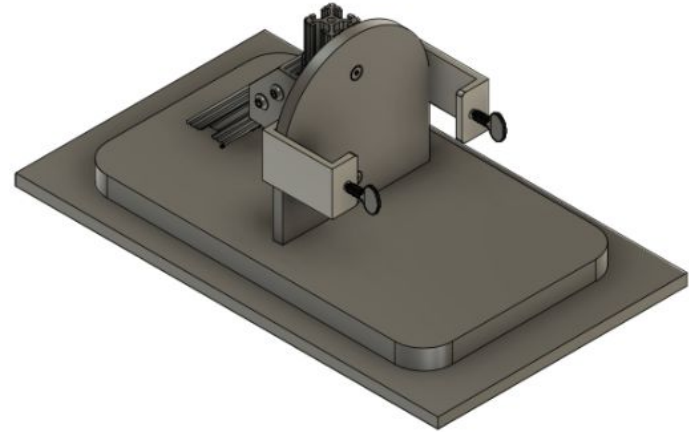


MA Design Recap

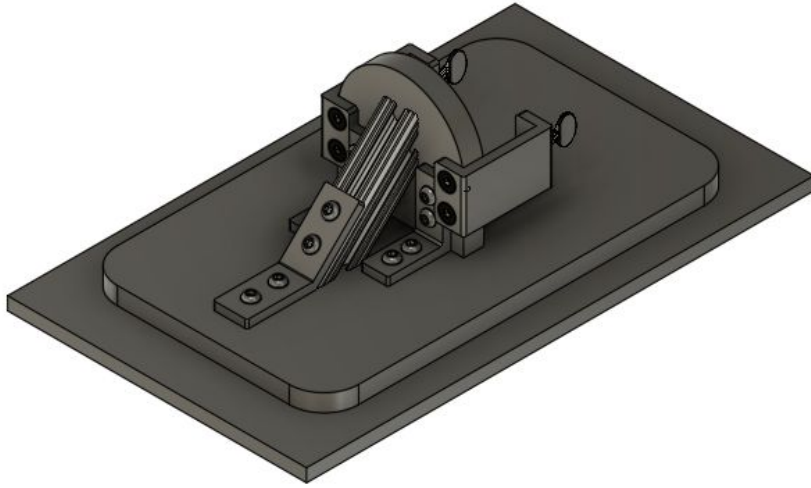
First Design



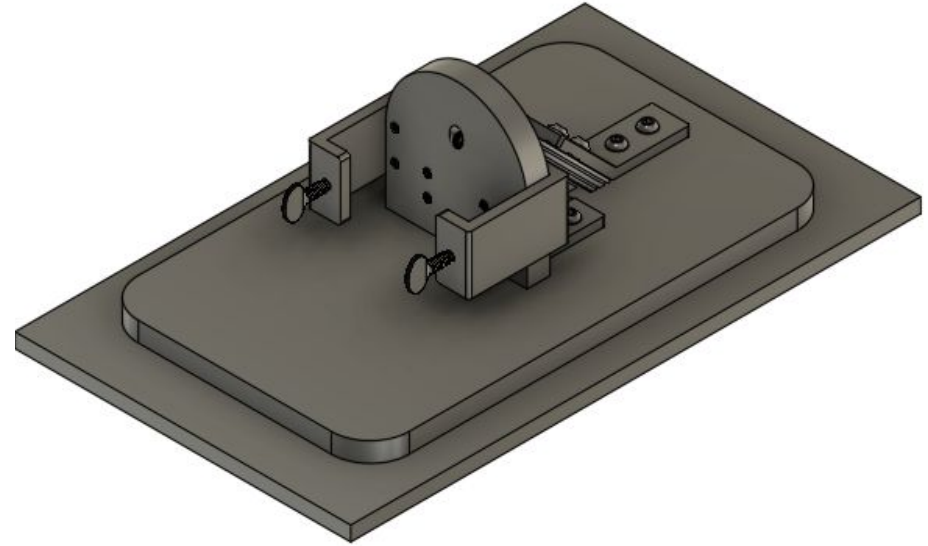
Second Design



Final MA Design



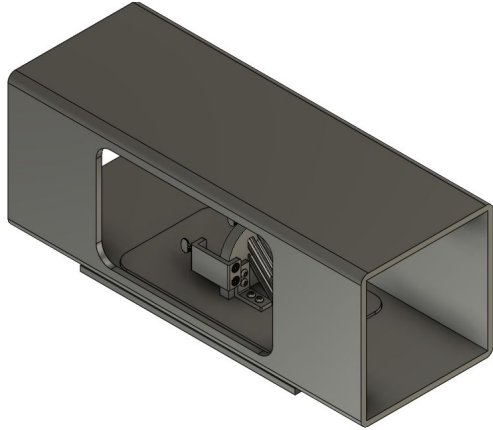
Auxiliary Rear View



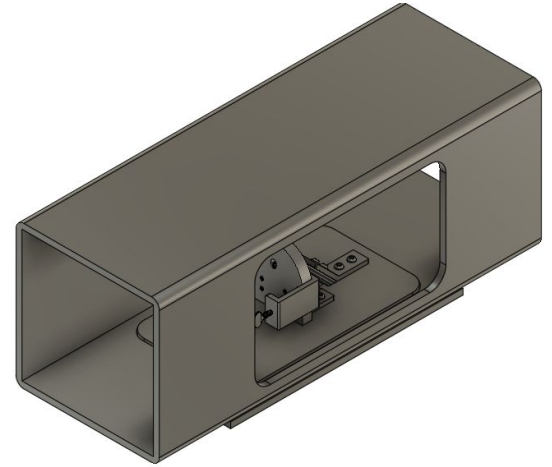
Auxiliary Front View

Final MA Design in Blast Tube

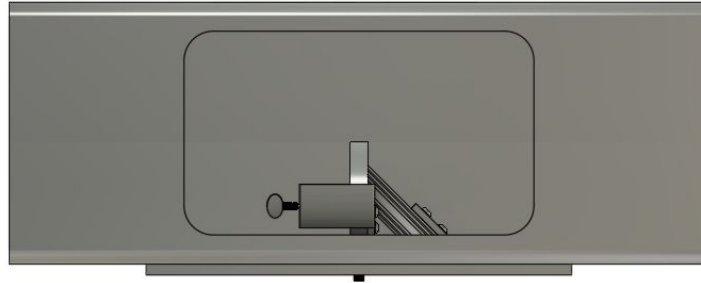
Auxiliary Rear View



Auxiliary Front View



Side View



Most Critical Tolerances

1. All Brackets

- a. **Features:** (1) Faces - Parallelism, (2) Angles - Angular Tolerance
- b. **Reason:** (1) An even bracket fit, (2) Proper angular support

2. Headboard and Baseboard

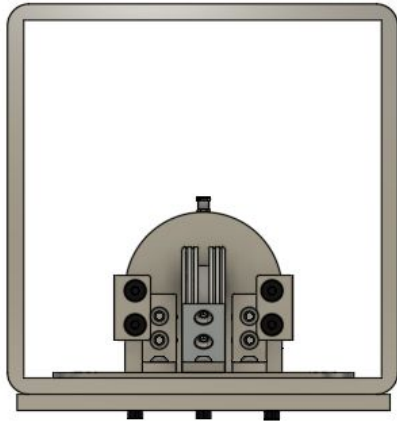
- a. **Features:** (1) Face - Flatness, (2) Threaded Hole Locations - Position Tolerance
- b. **Reason:** (1) Prevents gaps and stress, (2) Ensures thread alignment

Complete Assembly

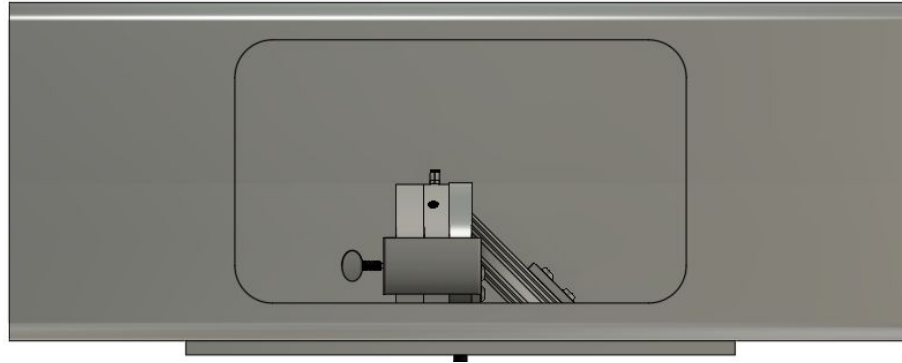
03

Updated Overall setup

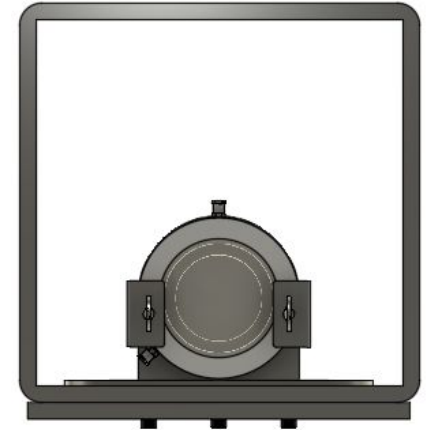
20% of space



*Left Side
View*

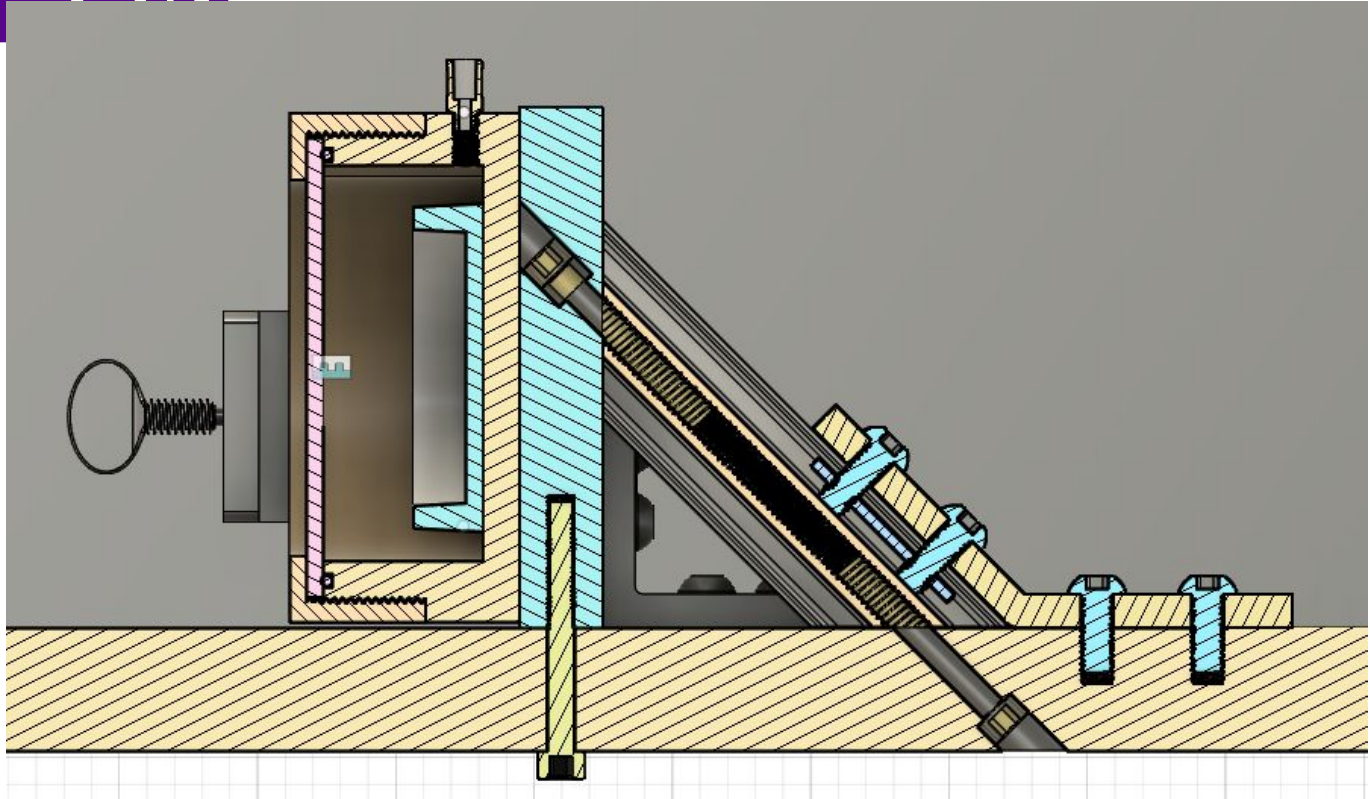


Length Wise View



*Right Side
View*

Cross-Section View - Full Assembly



Design Criteria

1. Area of the device is less than or equal 20% of the blast tube cross sectional area. ✓
2. Quick insertion of the PDMS substrate inside the apparatus (eliminate bolts). ✓
3. Elimination of all air bubbles. ✓
4. How the apparatus will be mounted inside the blast tube. ✓
5. Fitting the PDMS substrate inside a commercially available petri dish. ✓
6. Select Material for all you design components that are biocompatible and autoclavable. ✓
7. Pressure sensor integration. ✓

Questions?